IBD Digital Twin — system architecture

Five layers in one Python repo. Data flows one way: each layer reads only the layer above. Ground truth is used for labels and evaluation only.

1 Synthetic patients & data streams `src/generator/ → data/synthetic/`

800 virtual Indian IBD patients × 365 days, seeded (seed 42). Latent inflammation drives every stream with lags, noise, decoys (colds, stress, NSAIDs, infections) and missing days.

- EHR: demographics, Montreal class, ICD-10, NOD2 / IL23R / HLA
- Wearables: HRV, RHR, steps, SpO_2, sleep, skin temp, resp rate
- Daily symptom check-in (partial Mayo / HBI)
- Labs: calprotectin, CRP, Hb, ferritin…
- Sweat sensor (investigational)

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2 Fusion & personal baselines `src/features/`

One row per patient-day built only from data ≤ day t (tested). Each signal is compared with the patient's own robust baseline (median / MAD over observably quiet days).

- Deviation z-scores
- 7 / 14 / 28-day mean, SD, min, max, slope
- Co-movement count
- Lab last / delta / two-high flag
- Context flags (NSAID, antibiotic, cold, steroid taper)
- Inflammation burden

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3 Prediction, explanation & what-if `src/models/`

Will a flare start in the next 14 days? Logistic baseline vs LightGBM, 5-fold cross-validation grouped by patient, Platt-calibrated, alert threshold chosen on validation patients.

- 14-day risk (calibrated)
- TreeSHAP → top-5 plain-language drivers
- What-if: adherence, anti-TNF, stop NSAIDs (model-estimated)
- Ablation A / B / C

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4 Alerts & safety rules `src/alerts/`

Two tiers. Trend alerts over days to weeks; emergency patterns within hours. A rule-based Cancer Safety panel: surveillance support, not cancer detection.

- 🟠 Risk ≥ 20% → home calprotectin
- 🔴 Risk ≥ 50% → teleconsult
- 🚨 6 emergency rules
- CRC risk level, colonoscopy due date, red flags

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Ground truth — never a model input

events.csv (flares, emergencies, decoys), ground_truth/latent_daily (labels, latent inflammation), hidden_params.csv.

Used only to build the label **flare_within_14d** and to evaluate. Lab *reason / setting* columns are dropped. Only prescriptions and logged colds / infections are read from events.csv as context.

Traffic light

- 🟢 Low < 20%
- 🟠 Rising 20–50%
- 🔴 High ≥ 50%

Emergency rules

- Acute severe colitis
- VTE / PE
- Sepsis on steroids
- Bowel obstruction
- Stoma dehydration
- TB reactivation on anti-TNF

Design choices

- Disease-agnostic layers 2–5 (COPD, heart failure next)
- Sensor-agnostic: new streams are new columns

5 **Doctor dashboard** `src/dashboard/app.py (Streamlit)`

Population view sorted by risk, with emergency and overdue-surveillance lists. Patient view: card, risk gauge & trend, signals vs baseline band, labs, "why", what-if, suggested action, alerts, Cancer Safety, sweat (investigational). A date slider replays any day.

- Out-of-fold scoring in the UI

Stack: Python 3.11 · pandas · numpy · pyarrow · scikit-learn · LightGBM · Streamlit · Plotly · pytest

Synthetic data only · flags risk, never diagnoses